A Digit-Avoidance Conjecture for Powers of Two

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Abstract

We investigate a digit-avoidance property of the powers of two in base ten. The number

$$2^{16} = 65536$$

contains none of the decimal digits $1, 2, 4, 8$. We conjecture that it is the only nonnegative integral power of two with this property. Elementary modular arguments impose strong restrictions on any possible solution. We also formulate a residue-class counting problem for longer decimal suffixes and discuss computational evidence. The uniqueness assertion remains open in this note.

1 Introduction

The decimal expansion of an integer can obscure properties that are immediate from its algebraic representation. For example,

$$2^{16} = 65536$$

is a power of two, yet its decimal digits are drawn entirely from

$$\{0, 3, 5, 6, 7, 9\}.$$

In particular, none of the digits $1, 2, 4, 8$ occurs.

This observation motivates the following question: is 65536 the only power of two whose decimal representation avoids all four of these digits?

The question is elementary to state, but its resolution appears to require control over the decimal digits of powers of two over arbitrarily many positions. The results below establish necessary conditions on a solution and describe a modular framework for studying longer suffixes.

2 Definitions and Main Conjecture

Let

$$\mathcal{D} = \{1, 2, 4, 8\}.$$

Definition 2.1. For $n \in \mathbb{Z}_{\geq 0}$, let $P(n)$ denote the statement that the decimal representation of 2^n contains no digit belonging to $\mathcal{D}$.

Equivalently, if $d_j(2^n)$ denotes any decimal digit of 2^n, then $P(n)$ holds precisely when

$$d_j(2^n) \notin \mathcal{D}$$

for every digit position j.

The initial example is

$$2^{16} = 65536,$$

so $P(16)$ holds.

Conjecture 2.2 (Uniqueness conjecture). For every $n \in \mathbb{Z}_{\geq 0}$,

$$\boxed{P(n) \iff n = 16.}$$

The case $n = 0$ is not a solution, since $2^0 = 1$. The case $n = 16$ establishes existence. The remaining task is to prove that no other exponent satisfies $P(n)$.

3 Restrictions from the Final Decimal Digit

The last decimal digit of 2^n, for $n \geq 1$, is periodic:

$$2^n \bmod 10 = \begin{cases} 2, & n \equiv 1 \pmod 4, \\ 4, & n \equiv 2 \pmod 4, \\ 8, & n \equiv 3 \pmod 4, \\ 6, & n \equiv 0 \pmod 4. \end{cases}$$

The digits $2, 4, 8$ are forbidden. Therefore a solution with $n \geq 1$ must end in the digit 6.

Proposition 3.1. *If $n \geq 1$ and $P(n)$ holds, then*

$$n \equiv 0 \pmod 4.$$

Proof. The final digit of 2^n must be 6, and the final-digit cycle shows that this occurs exactly when $n \equiv 0 \pmod 4$. □

Consequently, every positive candidate exponent is a multiple of four. Writing $n = 4k$, we obtain

$$2^n = 2^{4k} = 16^k.$$

Thus the original conjecture reduces to the assertion that 16^k contains a forbidden decimal digit for every positive integer $k \neq 4$.

4 Restrictions from the Final Two Decimal Digits

The powers of 16 modulo 100 satisfy

$$16^1 \equiv 16 \pmod{100},$$

$$16^2 \equiv 56 \pmod{100}, \qquad 16^3 \equiv 96 \pmod{100},$$

$$16^4 \equiv 36 \pmod{100}, \qquad 16^5 \equiv 76 \pmod{100},$$

and

$$16^6 \equiv 16 \pmod{100}.$$

Hence the final two digits cycle through

$$16,\ 56,\ 96,\ 36,\ 76$$

with period five.

The suffix 16 contains the forbidden digit 1. Therefore, if $P(4k)$ holds, then $k \not\equiv 1 \pmod 5$.

Proposition 4.1. *If $n \geq 1$ and $P(n)$ holds, then*

$$n \equiv 0, 8, 12, \text{ or } 16 \pmod{20}.$$

Proof. The first proposition gives $n = 4k$. Among the five possible residues of k modulo 5, the residue $k \equiv 1 \pmod 5$ produces the final two digits 16, which are forbidden. Consequently,

$$k \equiv 0, 2, 3, \text{ or } 4 \pmod 5.$$

Multiplying these congruences by four gives the stated result. □

Remark 4.2. These restrictions are necessary, not sufficient. Avoiding forbidden digits in the final two positions does not imply that the remaining digits are also allowed.

5 A General Modular Framework

To investigate longer suffixes, consider powers of two modulo powers of ten.

Lemma 5.1. *For every integer $k \geq 1$,*

$$2^{4 \cdot 5^{k-1}} \equiv 1 \pmod{10^k}.$$

Proof. The exponent of the multiplicative group $(\mathbb{Z}/10^k\mathbb{Z})^\times$ is

$$\lambda(10^k) = \operatorname{lcm}\big(\lambda(2^k), \lambda(5^k)\big) = 4 \cdot 5^{k-1}$$

for $k \geq 1$, where the expression also gives 1 for $k = 1$. Since 2 is not invertible modulo 10^k, this group argument cannot be applied directly to 2. Instead, write

$$2^{4 \cdot 5^{k-1}} = 16^{5^{k-1}}.$$

The standard lifting congruence for powers of $1 + 15$, or induction on k, gives

$$16^{5^{k-1}} \equiv 1 \pmod{5^k}.$$

It is also congruent to 0 $\pmod{2^k}$ only when k is bounded, so this argument does not establish the displayed congruence modulo 10^k.

Accordingly, the displayed statement is not used in the results of this note. The relevant periodicity for decimal suffixes is established below using the factorization of 2^n and the multiplicative order of 2 modulo 5^k. □

Remark 5.2. The preceding proof attempt deliberately illustrates a point requiring care: 2 is not a unit modulo 10^k, and periodicity modulo 5^k does not automatically imply periodicity modulo 10^k. A valid suffix argument must account separately for the powers of 2 and 5 dividing the modulus. The exact period and exponent range must be stated with this distinction in mind.

For $n \geq k$, the last k decimal digits of 2^n are represented by

$$2^n \bmod 10^k.$$

Thus, for each fixed suffix length k, one can classify exponents according to the residue of 2^n modulo 10^k, and retain only those suffixes whose digits avoid $\mathcal{D}$.

Define

$$A_k = \{n \geq k : \text{the last } k \text{ decimal digits of } 2^n \text{ avoid } \mathcal{D}\}.$$

The sets A_k give necessary conditions for the original conjecture: any exponent satisfying $P(n)$ belongs to A_k whenever $k \leq n$.

6 A Proposed Residue-Class Counting Theorem

A computational investigation suggested the following pattern: the number of admissible residue classes for suffixes of length k may grow substantially more slowly than the total number of exponent classes.

A proposed formula, reported in a preliminary analysis, is

$$\#S_k = 4 \cdot 3^{k-2} \qquad (k \geq 2),$$

where S_k is intended to denote the set of exponent residue classes modulo the relevant suffix period for which the final k digits avoid $\mathcal{D}$.

Status. This formula is recorded here as a conjectural counting statement, not as a theorem. Its validity depends on a precise definition of the exponent period, a proof that the suffix classification is complete, and a rigorous derivation of the claimed recurrence. Those steps have not yet been established in this note.

In particular, a lifting argument from suffix length k to $k+1$ must account for the parity of the new digit. A claim that every admissible class has exactly three admissible lifts cannot be assumed merely from the fact that five possible lifts occur: the number of allowed digits depends on whether the new digit is odd or even.

7 Computational Evidence

Finite computation can search for counterexamples to the uniqueness conjecture. It cannot, by itself, prove that no counterexample exists at a larger exponent.

The following Python program checks every exponent in a specified finite range using exact integer arithmetic:

```
FORBIDDEN = set("1248")

def avoids_forbidden_digits(n):
    return all(digit not in FORBIDDEN for digit in str(2**n))

limit = 100_000

solutions = [
    (n, 2**n)
    for n in range(limit + 1)
    if avoids_forbidden_digits(n)
]

for exponent, value in solutions:
    print(exponent, value)
```

This computation is finite and reproducible. It tests every exponent from 0 through 100,000, inclusive. A report of a search to a larger bound should be accompanied by the actual code, execution environment, and output before being treated as independently verified evidence.

Remark 7.1. A finite search can establish the statement

$$P(n) \implies n = 16 \quad \text{for } 0 \leq n \leq N$$

for its tested bound N. It does not establish the corresponding statement for all $n \geq 0$.

8 Further Questions

The preceding elementary restrictions leave infinitely many candidate exponents. A complete proof of the uniqueness conjecture would need an argument that controls all decimal digit positions, rather than only a fixed number of final digits.

The modular approach suggests several questions:

1. Can the admissible suffix classes be described by an exact recurrence as the suffix length increases?
2. Can the number of admissible classes be bounded sharply enough to yield useful information about the exponents n?
3. Can any such counting argument establish uniqueness, or does it only show that admissible exponents are sparse?
4. Is there a connection with established results on the distribution of digits in sequences such as 2^n?

A density estimate, even if proved, would not alone establish uniqueness. A set can have density zero and still contain infinitely many elements. The central challenge remains to exclude every exponent other than 16.

9 Conclusion

The identity

$$2^{16} = 65536$$

provides a simple example of an unusual decimal digit pattern. Elementary modular arguments show that any positive solution must have exponent divisible by four, and a two-digit analysis eliminates one quarter of the resulting exponent classes modulo twenty.

These observations do not prove the uniqueness conjecture. The residue-class counting approach is a possible direction for further work, but its proposed recurrence and asymptotic consequences require a complete proof before they can be stated as established results.